

Ex. 9.8 E

Q31  $e^x \sin x$

Need extra terms

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$e^x \sin x = x + x^2 + \frac{1}{3}x^3 - \frac{1}{30}x^5 + \dots$$

| $e^x \sin x$    | 0 | 1 | 0 | $-\frac{1}{3!}$ | 0 | $\frac{1}{5!}$ | 0 | $-\frac{1}{7!}$ |
|-----------------|---|---|---|-----------------|---|----------------|---|-----------------|
| $\frac{1}{4!}$  |   |   |   |                 |   |                |   |                 |
| $-\frac{1}{5!}$ |   |   |   |                 |   |                |   |                 |
| $\frac{1}{2}$   |   |   |   |                 |   |                |   |                 |
| $-\frac{1}{3!}$ |   |   |   |                 |   |                |   |                 |

| $e^x \backslash \sin x$ | 0 | 1              | 0 | $-\frac{1}{3!}$          | 0 | $\frac{1}{5!}$          | 0 | $-\frac{1}{7!}$          |
|-------------------------|---|----------------|---|--------------------------|---|-------------------------|---|--------------------------|
| 1                       | 0 | 1              | 0 | $-\frac{1}{3!}$          | 0 | $\frac{1}{5!}$          | 0 | $-\frac{1}{7!}$          |
| 1                       | 0 | 1              | 0 | $-\frac{1}{3!}$          | 0 | $\frac{1}{5!}$          | 0 | $-\frac{1}{7!}$          |
| $\frac{1}{2!}$          | 0 | $\frac{1}{2}$  | 0 | $-\frac{1}{3! \cdot 2}$  | 0 | $\frac{1}{5! \cdot 2}$  | 0 | $-\frac{1}{7! \cdot 2}$  |
| $\frac{1}{8!}$          | 0 | $\frac{1}{3!}$ | 0 | $-\frac{1}{3! \cdot 3!}$ | 0 | $\frac{1}{3! \cdot 5!}$ | 0 | $-\frac{1}{7! \cdot 3!}$ |
| $\frac{1}{4!}$          | 0 | $\frac{1}{4!}$ | 0 | $-\frac{1}{3! \cdot 4!}$ | 0 | $\frac{1}{4! \cdot 5!}$ | 0 | $-\frac{1}{7! \cdot 4!}$ |
| $\frac{1}{8!}$          | 0 | $\frac{1}{3!}$ | 0 |                          | 0 |                         | 0 |                          |

$$e^x \sin x = x + x^2 + \left(\frac{1}{2} - \frac{1}{3!}\right)x^3 + \left(\frac{1}{3!} - \frac{1}{3!}\right)x^4 + \left(\frac{1}{4!} - \frac{1}{3! \cdot 2} + \frac{1}{5!}\right)x^5 + \dots$$

Q32

$$\ln(1+x) = x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{4}x^4$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$\frac{\ln(1+x)}{1-x} = x + \frac{1}{2}x^2 + \frac{5}{6}x^3 + \frac{7}{12}x^4 + \dots$$

Q29  $\frac{x}{3} \ln(1+x^2) \rightarrow \ln(1+u) \quad u=x^2$

Idea: Substitution

Derivative  
PDF ← Factorial  
power of  $x$

$$f(x) = \cancel{f(0)} + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!} + \dots \quad [\text{Mac.}]$$

$$f(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots$$